

Kevin Kuang

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EDUCATION

University of California, San Diego

San Diego, CA

Bachelor of Science in Mathematics-Computer Science, Minor in Computer Engineering

Sept. 2021 – June 2025

PROJECTS

TritonTube | *Go, gRPC, SQLite, ffmpeg, etcd, AWS*

May 2025 – June 2025

- Developed a distributed video hosting platform allowing users to upload content, using ffmpeg to transcode videos into adaptive bitrate formats for improved playback quality under fluctuating network conditions
- Implemented a dual metadata layer with SQLite for lightweight testing and etcd for production-scale deployment, enabling high availability and ensuring all services operate on the most up-to-date state
- Designed a consistent hashing algorithm for file distribution across storage nodes, ensuring even load balancing and seamless video accessibility during dynamic node scaling and recovery
- Built an admin CLI with gRPC support to manage the storage cluster by adding or removing nodes, automatically redistributing video files to maintain system consistency and prevent misrouted file requests

Data Center Simulation | *Go, Docker, Kubernetes, gRPC, wrk, Git, GCP*

Jan 2025 – March 2025

- Deployed a containerized restaurant microservices application using Kubernetes and gRPC, optimizing request handling by scaling up the bottlenecked pods to improve system throughput and minimize latency by 73%
- Identified bottlenecks using wrk, testing performance pre- and post-implementation to compare latency
- Enhanced system scalability by integrating persistent storage layers and implementing caching as a microservice, reducing overall response times by 70% while balancing overhead from cache invalidation
- Reduced microservice response latency by 63.9% with LRU caching under Zipfian access patterns

Embedded Visual Aid | *C, Python, Bash, Ollama, Git*

Jan 2025 – March 2025

- Developed a real-time accessibility system for visually impaired users, integrating 8 proximity sensors with haptic feedback and a laser-temperature sensor for auditory alerts
- Implemented an Earliest Deadline First (EDF) scheduler at the kernel level to prioritize critical proximity and temperature sensors, reduced response time of sensor and actuator activation time by 18%
- Designed a camera system with button-activated image capture, leveraging Llava on Ollama for local AI-driven analysis and text-to-speech conversion, reducing overall costs by approximately \$60 through local server processing
- Worked on optimizing real-time scheduling for task cancellation and reducing energy usage by 18% under varying load conditions

Developer's Journal | *HTML, CSS, JavaScript, Figma, Jest, Git*

March 2024 – June 2024

- Developed a CRUD-based Developer's Journal that implements code snippets, markdown-based documentation, and task/event management
- Managed a team of 9, overseeing the agile workflow and facilitating stand-ups to guarantee timely project delivery
- Configured a CI/CD pipeline with GitHub Actions to automate testing and deployment for continuous integration
- Designed the application with a local-first approach, ensuring offline functionality and data integrity
- Implemented 20+ unit tests with Jest and end-to-end testing with Puppeteer to validate functionality

CodePath - Cybersecurity Analysis | *Wireshark, Snort, Splunk, Python, Catalyst*

April 2024 – June 2024

- Investigated an AWS Cloud Bank Breach, identifying unauthorized access where an attacker escalated privileges to an administrative role and attempted to list and extract objects from an S3 bucket
- Conducted network threat analysis by detecting malicious activity patterns in timestamps, IP addresses, and request types within JSON logs through using signature-based detection with Splunk
- Revised Amazon's Data Access Playbook by restricting high-privilege roles and implementing time-limited session tokens to prevent unauthorized access
- Educated 15+ young professionals in threat modeling, defensive proactive strategies, and best practices in password security at a corporate security workshop, preparing them to better secure critical assets

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, HTML/CSS, Go, SQL

Frameworks/Libraries: React, Node.js, Jest, JUnit, Pandas, NumPy, Matplotlib

Developer Tools: Git, Docker, Google Cloud Console, Amazon Web Services, Kubernetes, etcd, GNU Debugger, Vim, Splunk, LaTeX